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# =====
# 📊 EXPLORATORY DATA ANALYSIS
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# =====
# 0. UPLOAD FILE (COLAB STEP)
# =====
from google.colab import files
uploaded = files.upload()

# =====
# 1. LOAD DATA
# =====
df = pd.read_csv("netflix_titles.csv")

print("\n--- FIRST 5 ROWS ---")
print(df.head())

# =====
# 2. BASIC INFO
# =====
print("\n--- SHAPE ---")
print(df.shape)

print("\n--- INFO ---")
print(df.info())

print("\n--- STATISTICS ---")
print(df.describe(include='all'))

# =====
# 3. MISSING VALUES CHECK
# =====
print("\n--- MISSING VALUES ---")
print(df.isnull().sum())

# Fill numeric missing values with median
df = df.fillna(df.median(numeric_only=True))

# Fill categorical missing values with mode
for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].fillna(df[col].mode()[0])

# =====
# 4. DUPLICATES
# =====
print("\n--- DUPLICATES ---")
print(df.duplicated().sum())

df = df.drop_duplicates()

# =====
# 5. COLUMN NAMES
# =====
print("\n--- COLUMNS ---")
print(df.columns)

# =====
# 6. CORRELATION HEATMAP
# =====
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()

# =====
# 7. HISTOGRAMS
# =====
df.hist(figsize=(12,10))
plt.suptitle("Feature Distributions")
plt.show()

# =====
# 8. BOX PLOTS
# =====
plt.figure(figsize=(12,6))

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plt.figure(figsize=(12,10))
sns.boxplot(data=df.select_dtypes(include=np.number))
plt.xticks(rotation=90)
plt.title("Outlier Detection")
plt.show()

# =====
# 9. PAIRPLOT (NUMERIC ONLY)
# =====
sns.pairplot(df.select_dtypes(include=np.number))
plt.show()

# =====
# 10. CATEGORY ANALYSIS
# =====
cat_cols = df.select_dtypes(include="object").columns

for col in cat_cols:
    print(f"\n--- VALUE COUNTS: {col} ---")
    print(df[col].value_counts().head(10))

    plt.figure(figsize=(8,4))
    sns.countplot(y=df[col], order=df[col].value_counts().index[:10])
    plt.title(f"Top 10 values of {col}")
    plt.show()

# =====
# 11. SAVE CLEANED DATA
# =====
df.to_csv("cleaned_netflix_titles.csv", index=False)

print("\n✅ EDA COMPLETE. Cleaned dataset saved as 'cleaned_netflix_titles.csv'")
```


Choose files netflix_titles.csv

netflix_titles.csv(text/csv) - 3408471 bytes, last modified: 06/06/2026 - 100% done
 Saving netflix_titles.csv to netflix_titles (1).csv

--- FIRST 5 ROWS ---

	show_id	type	title	director	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

	cast	country	\
0	NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	
3	NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	

	date_added	release_year	rating	duration	\
0	September 25, 2021	2020	PG-13	90 min	
1	September 24, 2021	2021	TV-MA	2 Seasons	
2	September 24, 2021	2021	TV-MA	1 Season	
3	September 24, 2021	2021	TV-MA	1 Season	
4	September 24, 2021	2021	TV-MA	2 Seasons	

	listed_in	\
0	Documentaries	
1	International TV Shows, TV Dramas, TV Mysteries	
2	Crime TV Shows, International TV Shows, TV Act...	
3	Docuseries, Reality TV	
4	International TV Shows, Romantic TV Shows, TV ...	

	description
0	As her father nears the end of his life, filmm...
1	After crossing paths at a party, a Cape Town t...
2	To protect his family from a powerful drug lor...
3	Feuds, flirtations and toilet talk go down amo...
4	In a city of coaching centers known to train I...

--- SHAPE ---
 (8807, 12)

--- INFO ---

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 8807 entries, 0 to 8806

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8803 non-null	object
9	duration	8804 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

dtypes: int64(1), object(11)

memory usage: 825.8+ KB

None

--- STATISTICS ---

	show_id	type	title	director	cast	\
count	8807	8807	8807	6173	7982	
unique	8807	2	8804	4528	7692	
top	s8807	Movie	Feb-09	Rajiv Chilaka	David Attenborough	
freq	1	6131	2	19	19	
mean	NaN	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	NaN	

	country	date_added	release_year	rating	duration	\
count	7976	8797	8807.000000	8803	8804	
unique	748	1767	NaN	17	220	
top	United States	January 1, 2020	NaN	TV-MA	1 Season	
freq	2818	109	NaN	3207	1793	
mean	NaN	NaN	2014.180198	NaN	NaN	
std	NaN	NaN	8.819312	NaN	NaN	
min	NaN	NaN	1925.000000	NaN	NaN	
25%	NaN	NaN	2013.000000	NaN	NaN	
50%	NaN	NaN	2017.000000	NaN	NaN	
75%	NaN	NaN	2019.000000	NaN	NaN	

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max          NaN          NaN  2021.000000  NaN          NaN

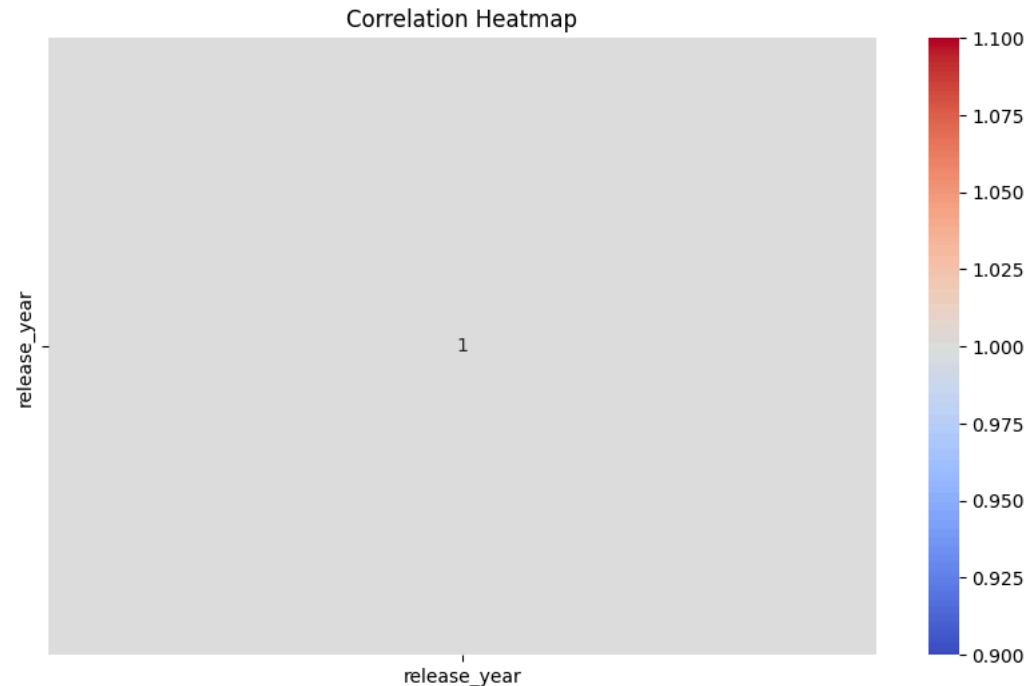
count          listed_in  \
unique          8807
top          Dramas, International Movies
freq          362
mean          NaN
std          NaN
min          NaN
25%          NaN
50%          NaN
75%          NaN
max          NaN

count          description
unique          8807
top          Paranormal activity at a lush, abandoned prope...
freq          4
mean          NaN
std          NaN
min          NaN
25%          NaN
50%          NaN
75%          NaN
max          NaN

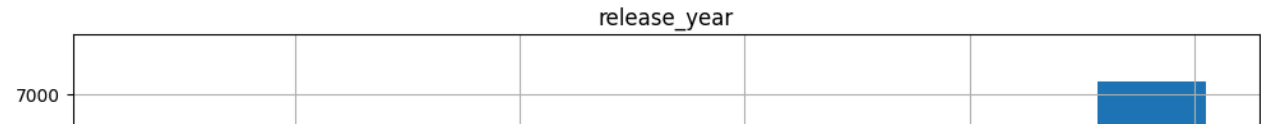
--- MISSING VALUES ---
show_id      0
type         0
title        0
director     2634
cast         825
country      831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description   0
dtype: int64

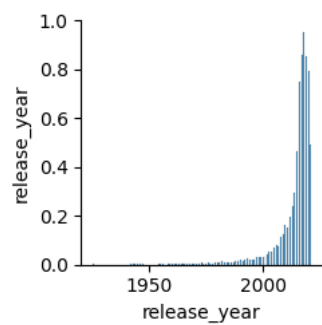
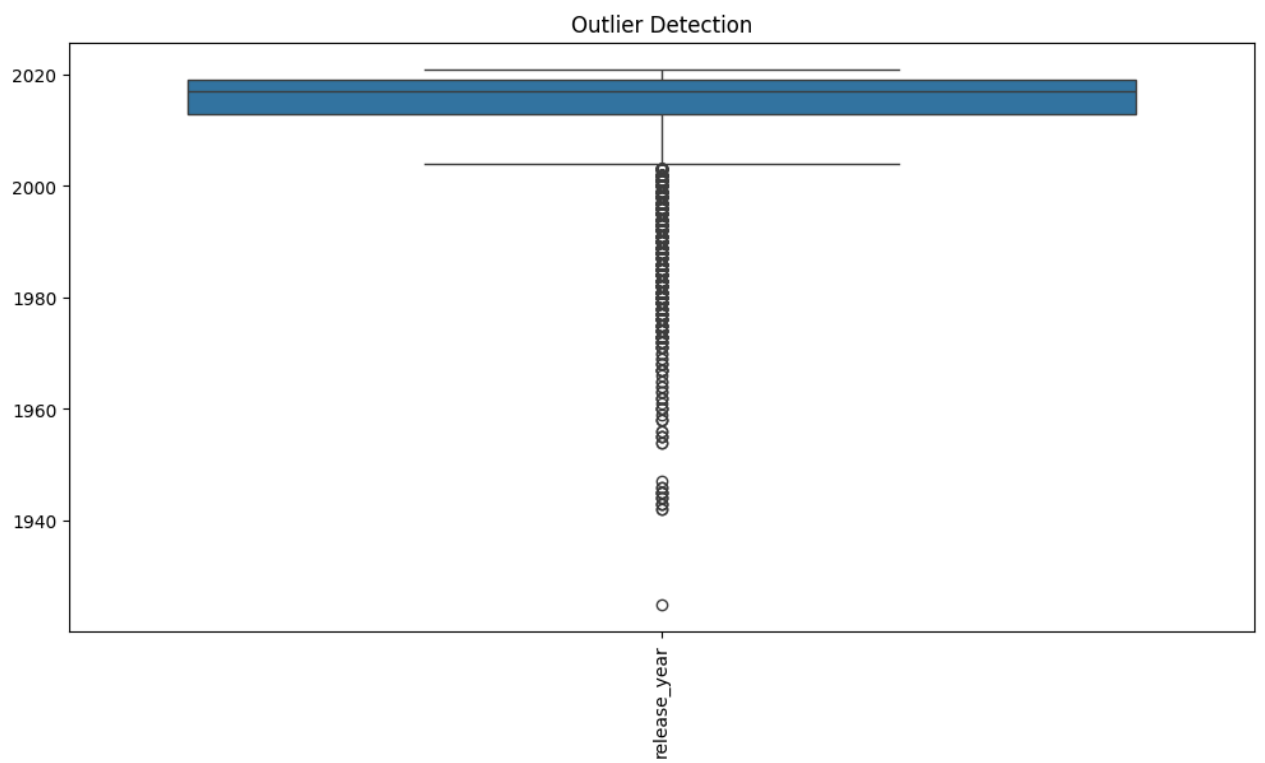
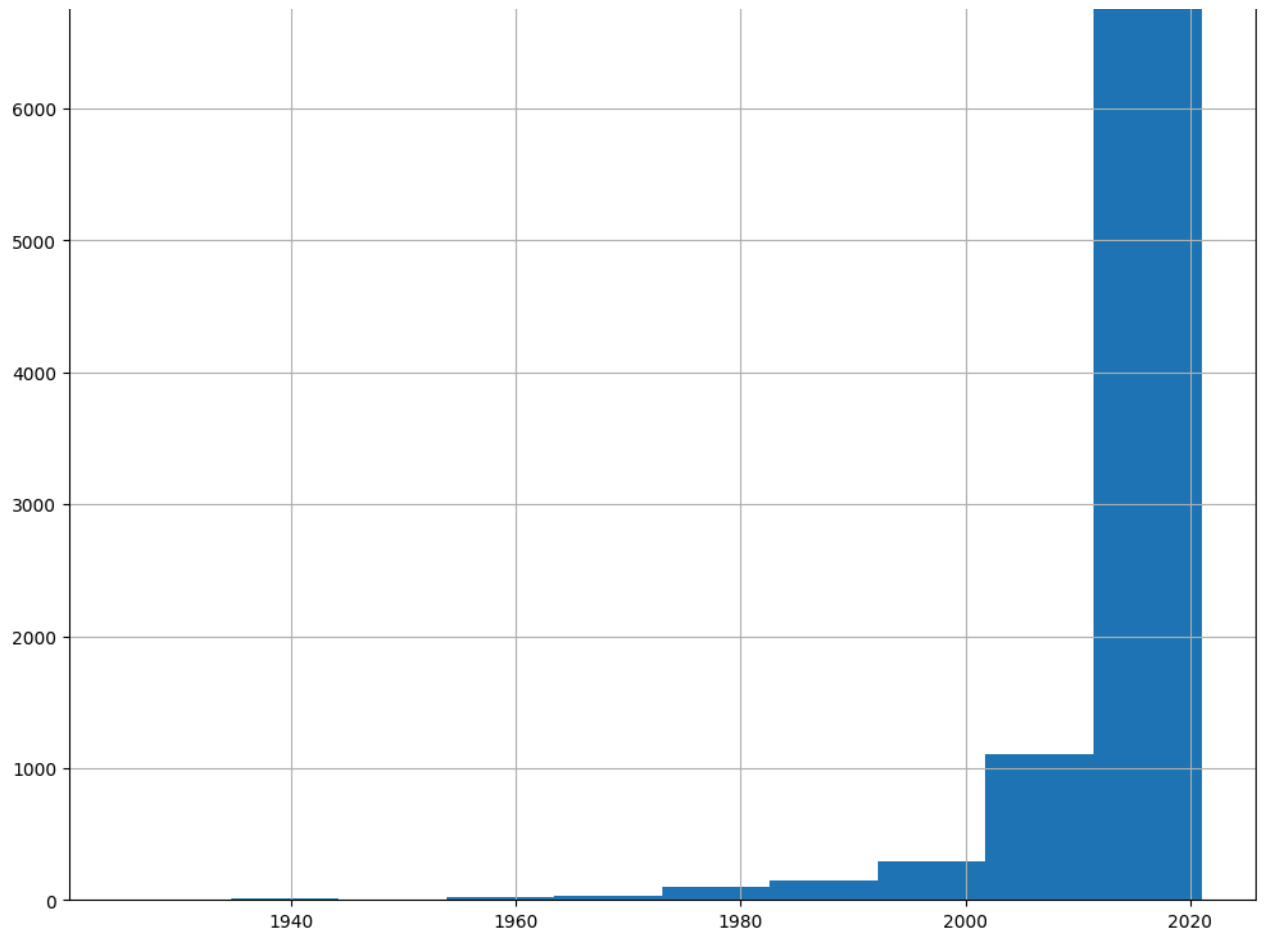
--- DUPLICATES ---
0

--- COLUMNS ---
Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
      'release_year', 'rating', 'duration', 'listed_in', 'description'],
      dtype='object')
```



Feature Distributions



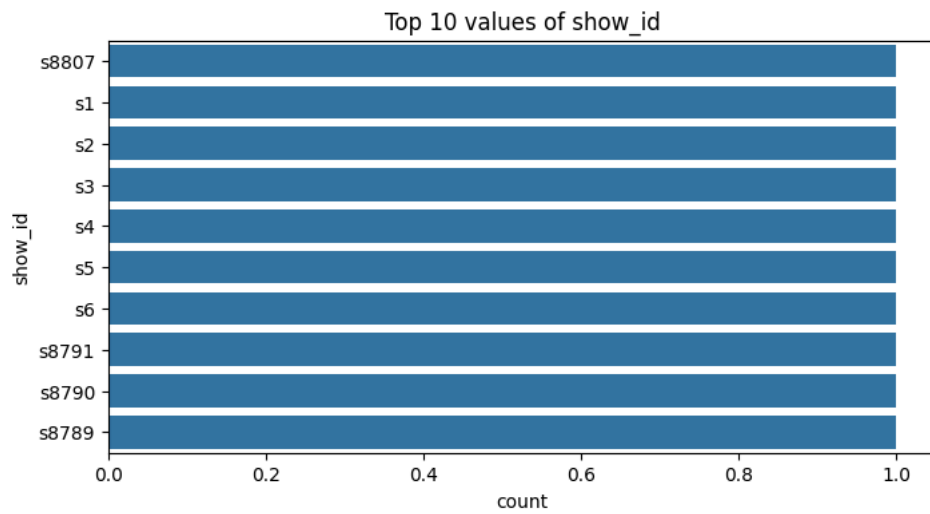


--- VALUE COUNTS: show_id ---
show_id

```

show_id
s8807    1
s1       1
s2       1
s3       1
s4       1
s5       1
s6       1
s8791    1
s8790    1
s8789    1
Name: count, dtype: int64

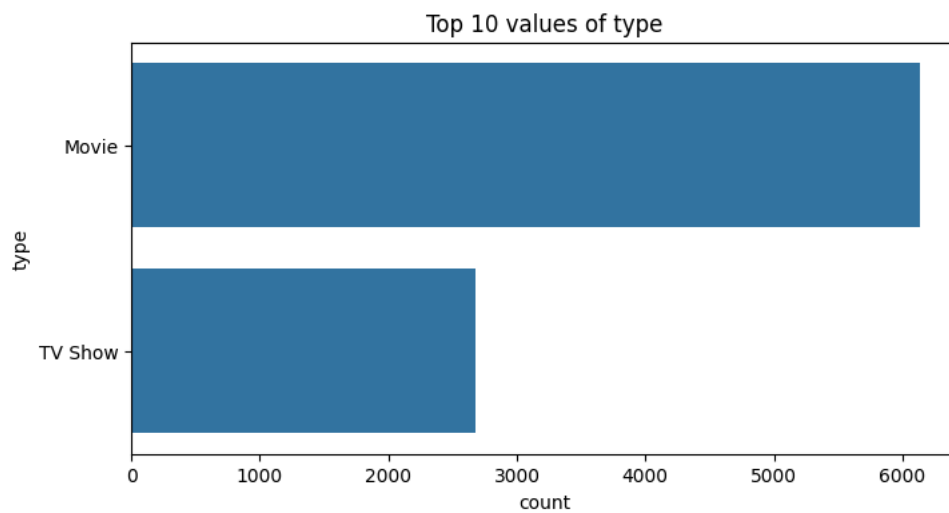
```



```

--- VALUE COUNTS: type ---
type
Movie      6131
TV Show    2676
Name: count, dtype: int64

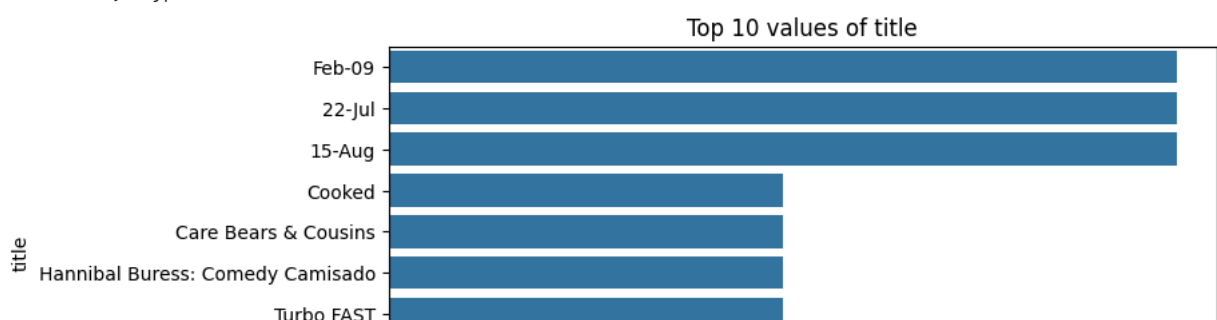
```

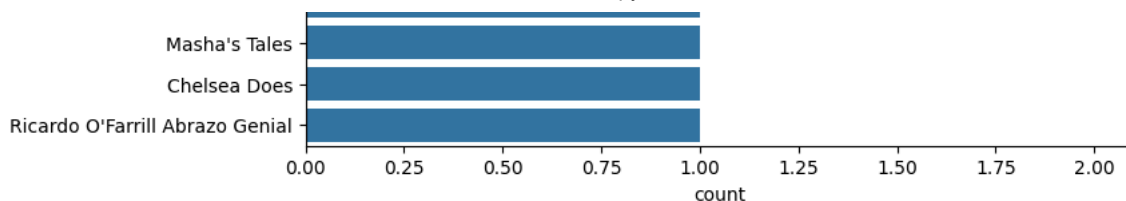


```

--- VALUE COUNTS: title ---
title
Feb-09      2
22-Jul      2
15-Aug      2
Cooked      1
Care Bears & Cousins  1
Hannibal Buress: Comedy Camisado  1
Turbo FAST  1
Masha's Tales  1
Chelsea Does  1
Ricardo O'Farrill Abrazo Genial  1
Name: count, dtype: int64

```

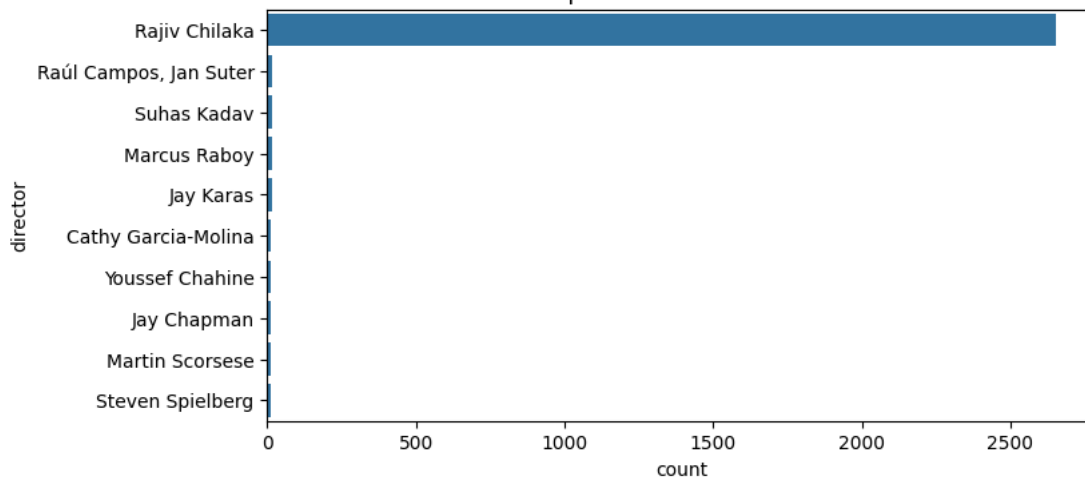




--- VALUE COUNTS: director ---

```
director
Rajiv Chilaka      2653
Raúl Campos, Jan Suter    18
Suhas Kadav        16
Marcus Raboy       16
Jay Karas          14
Cathy Garcia-Molina 13
Youssef Chahine    12
Jay Chapman        12
Martin Scorsese    12
Steven Spielberg   11
Name: count, dtype: int64
```

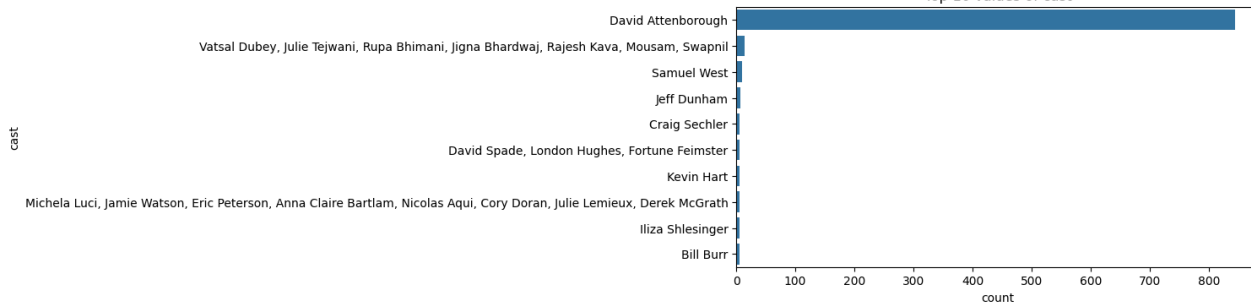
Top 10 values of director



--- VALUE COUNTS: cast ---

```
cast
David Attenborough      84
Vatsal Dubey, Julie Tejwani, Rupa Bhimani, Jigna Bhardwaj, Rajesh Kava, Mousam, Swapnil 1
Samuel West             1
Jeff Dunham
Craig Sechler
David Spade, London Hughes, Fortune Feimster
Kevin Hart
Michela Luci, Jamie Watson, Eric Peterson, Anna Claire Bartlam, Nicolas Aquí, Cory Doran, Julie Lemieux, Derek McGrath
Iliza Shlesinger
Bill Burr
Name: count, dtype: int64
```

Top 10 values of cast



--- VALUE COUNTS: country ---

```
country
United States      3649
India              972
United Kingdom     419
Japan              245
South Korea        199
Canada             181
Spain              145
France             124
Mexico             110
Egypt              106
Name: count, dtype: int64
```

Top 10 values of country

